



SHORT REPORT

Diagnostic yield of endoscopic ultrasound–guided fine-needle aspiration of tubercular lymphadenitis using combination of cytology and Gene Xpert *Mycobacterium tuberculosis*/rifampicin (MTB/RIF) genes

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Received: 23 November 2020 / Accepted: 3 December 2020
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Abstract

Diagnostic yield of an automated molecular test, Gene Xpert *Mycobacterium tuberculosis*/rifampicin (MTB/RIF), was evaluated in this study to simultaneously detect the MTB gene and resistance to rifampicin (RIF) on cytology samples acquired via endoscopic ultrasound (EUS)–guided fine-needle aspiration (FNAC) in suspected tubercular lymphadenitis. Microscopy, cytology, Gene Xpert MTB/RIF assay data on Acid-fast bacillus (AFB), and traditional culture of lymph nodes were retrospectively analyzed. Thirty-one patients (median age 33.5 years, inter-quartile range [IQR] 21–66, 18, 58% female) presented with fever (28, 90%), dysphagia (2, 7%), and recurrent subacute intestinal obstruction (1, 3%). Gene Xpert showed higher sensitivity (30, 97%) compared to the other tests: cytology (23, 77%; odds ratio [OR] 8.8, 95% confidence interval [CI] 1.0–76.9; $p = 0.05$), AFB smears (12, 39%; OR 50, 95% CI 5.9–420.4; $p = 0.00001$), and conventional culture (4, 13%; OR 188.5, 95% CI 19.7–1796.3; $p = 0.0000$). We conclude that Gene Xpert MTB/RIF test on EUS-guided FNAC samples is very useful to diagnose tubercular lymphadenitis.

Keywords Acid-fast bacillus · Acquired immunodeficiency syndrome · Anti-tubercular treatment · Drug-resistant tuberculosis · Mediastinum · Pyrexia · Tuberculosis · Ziehl-Neelsen (ZN) stain

Introduction

Tuberculosis (TB) is a significant public health problem in developing countries, including India [1]. Lymphadenitis is the most common form of extra-pulmonary TB [2]. Morbidity in patients with tubercular lymphadenitis may be reduced by early diagnosis and treatment. Mediastinal or intra-abdominal lymphadenopathy has many differential diagnoses;

hence, laboratory confirmation of the exact etiology is essential to guide therapy.

The initial diagnostic tools for tubercular lymphadenitis and extra-pulmonary TB are cytology and conventional smear microscopy [3, 4]. However, these tests' limitations are low specificity and sensitivity due to the pauci-bacillary nature of lymph nodal fine needle aspirates (FNA) [3, 5, 6]. Therefore, there is a need for more reliable techniques. Gene Xpert

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Bullet points of the study highlights

What is already known?

- Gene Xpert, a real-time polymerase chain reaction (PCR), is useful on sputum samples for the diagnosis of acquired immunodeficiency syndrome-associated tuberculosis (TB) and rifampicin resistance.

What is new in this study?

- Gene Xpert, a reliable and easily applicable method in endoscopic ultrasound (EUS)-guided TB lymph nodal aspirates with the potential to significantly improve rapid diagnosis TB lymphadenitis.

What are the future clinical and research implications of the study findings?

- Gene Xpert is able to provide rapid diagnosis of lymph nodal tuberculosis by EUS-guided fine needle aspiration, which has major clinical relevance.
- The results open the scope for larger studies.

Mycobacterium tuberculosis/rifampicin (MTB/RIF) (Cepheid, Sunnyvale, USA) has been recommended for the diagnosis of pulmonary TB by the World Health Organization (WHO) in 2010 [7]. The Gene Xpert assay consists of a closed system based on real-time polymerase chain reaction (PCR). It requires very little technical expertise, coupled with the advantage of rapid results (within 2 h), along with simultaneous detection of rifampicin resistance [8, 9].

Gene Xpert has been validated for use on sputum samples, in patients with acquired immunodeficiency syndrome-associated TB and multidrug-resistant infection [9–13]. WHO strongly recommends its widespread use for the diagnosis of pulmonary TB. Recent studies have evaluated this assay's usefulness on non-respiratory samples in patients with extra-pulmonary TB [14, 15].

In 2014, WHO recommended the use of Gene Xpert over the conventional method (i.e. microscopy, culture, or histopathology) for testing non-respiratory specimens (lymph nodes and other tissues) in suspected extra-pulmonary TB [8]. We evaluated the utility of Gene Xpert for diagnosing mediastinal and intra-abdominal tubercular lymph nodes, on routinely collected endoscopic ultrasonography (EUS)-guided FNA samples and compared it with cytology, smear microscopy, and culture.

Methods

Retrospective analysis of prospectively maintained hospital records of patients with mediastinal and intra-abdominal lymphadenopathy whose EUS-guided FNA had been done

over an eleven-month period (between May 2017 and October 2018) and had been diagnosed as TB was undertaken.

Collection of data on demographic, clinical, and laboratory parameters

Data were obtained from the electronic hospital information system (HIS), and case files; these included information regarding clinical presentation, demography, investigations, procedure details, anti-tubercular therapy (ATT) and its duration, and follow-up. Follow-up data were obtained by physical outpatient visits or telephonic interviews and included information about improvement in symptoms, ATT-induced toxicity, and liver function tests.

The technique of EUS and sonographic characteristics of the lymph nodes

EUS was performed using linear array echo-endoscope (GF UCT 180; Olympus Ltd., Tokyo, Japan) under sedation using midazolam and pentazocine injections by three endoscopists (SM, HKN, and UCG) after informed consent. EUS evaluation was done from the mediastinum, gastroesophageal junction, duodenal bulb, and second part of the duodenum to screen subcarinal, aortopulmonary, celiac, periportal, peripancreatic, and aortocaval lymph nodes. The suitability of most appropriate lymph nodes for FNA was based on size, shape, location, and echotexture. For sampling, a large lymph node was preferred over small; homogenous preferred over the heterogeneous node; those with hypoechoic center were preferred. The puncture was made using a 22-gauge needle (Expect FNA needle, Boston Scientific, USA or EchoTip

Ultra needle, Cook Medical, Bloomington, IN, USA) into the deepest part of the lymph node, the stylet was removed entirely, tissue acquisition was done by fanning technique with multiple passes with the standard technique of 5–10 to and fro movements, and the aspirated material was taken using a 10-mL syringe, attached to the hub of the needle, followed by application of negative suction and very slow withdrawal of the needle (Fig. 1). EUS aspirates were expelled from the needle by re-insertion of the stylet. In all the cases, dry slides were sent for cytopathology, and AFB staining and aspirates were sent for Gene Xpert analysis and conventional culture for *Mycobacterium tuberculosis*. There was no onsite pathologist for the study. Samples considered adequate by visual assessment of slides were included.

Data on EUS characteristics of the lymph nodes and procedure (size, shape, location, number, borders, echotexture, calcification, number of needle passes made, and procedure-related complications), cytology, AFB microscopy, Gene Xpert for MTB/RIF assay, and conventional culture were extracted. Diagnostic accuracy (sensitivity and false negative rate) was calculated with respect to clinical response to ATT (considered as the gold standard).

Cytological and microbiological evaluation

Grading of cytology slides was done as follows: I—epithelioid granulomatous reaction with caseation; II—epithelioid granulomatous reaction without caseation; III—non-granulomatous reaction with necrosis; IV—nonspecific changes; V—inadequate sample. The specimens positive for TB were graded as I–III [3].

Xpert MTB/RIF platform (Cepheid, Sunnyvale, CA, USA) detects *Mycobacterium tuberculosis* DNA directly from samples and rifampicin resistance by beta subunit of RNA

polymerase (rpo B) gene mutation; the residual deposit was decontaminated and processed using conventional microbiological techniques of microscopy with auramine-phenol and Ziehl-Neelsen (ZN) staining and liquid culture (BACTEC MGIT 960 Systems; Becton Dickinson, Oxford, UK). A manual step was mixing 1.5 mL bactericidal sample-reagent with 0.5 mL of FNA aspirated sample to obtain a total volume of 2 mL. The mixture was incubated at room temperature for 15 min before transferring into the Gene Xpert test cartridge. Then, the cartridges were placed in Xpert device and results of *Mycobacterium tuberculosis* or/and RIF resistance were interpreted as detected, not detected, and indeterminate. The test was interpreted within 2 h using a disposable cartridge. The test was repeated if the result showed an error. Definite TB was defined as positive cytology (grade I–III) and or positive AFB smear and or detection of *Mycobacterium tuberculosis* by Gene Xpert test with consistent clinical features.

Diagnosis of tuberculosis

Diagnosis of TB was confirmed by documenting clinical response to ATT (resolution of fever, abdominal pain, increase in appetite, weight gain etc.), during a follow-up period of 3 months or more.

Statistical analysis

Continuous data were expressed as mean $\pm$ standard deviation or median and inter-quartile range depending on distribution while categorical data as frequency and percentages. Measures of diagnostic accuracy were calculated in terms of sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and overall accuracy (OA). A

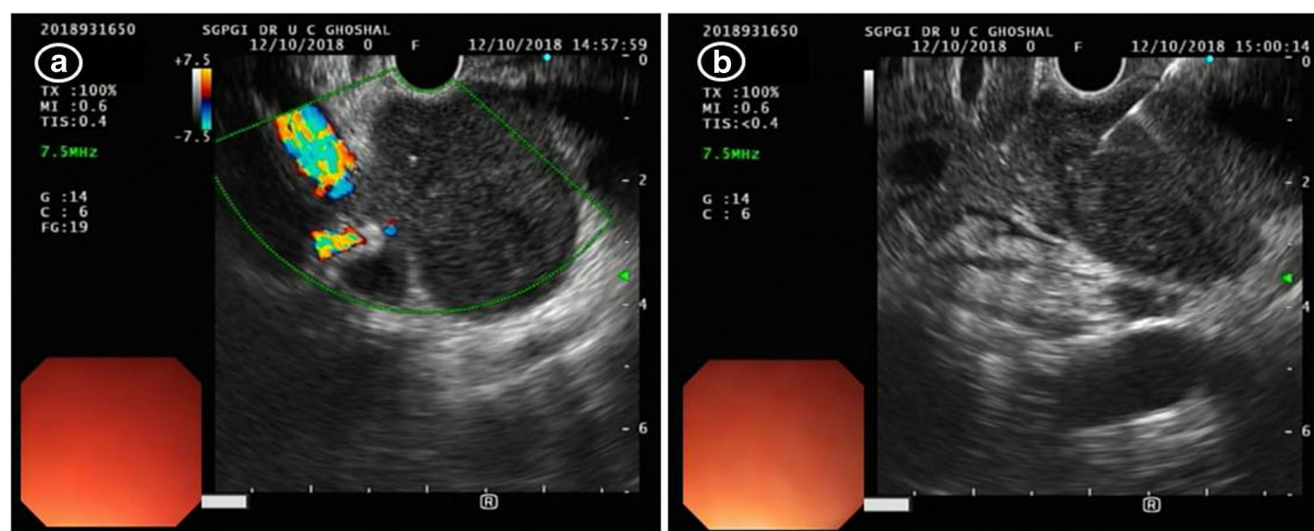
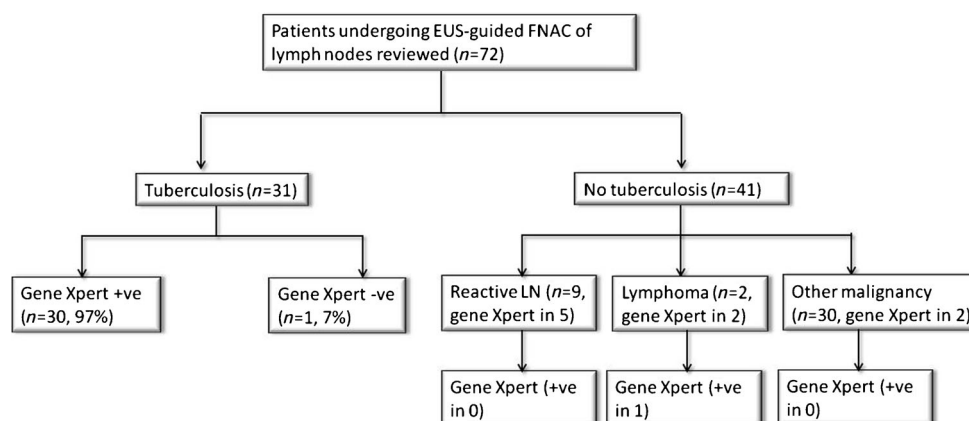


Fig. 1 **a** Linear endoscopic ultrasound (EUS) examination demonstrated a large hypoechoic peripancreatic lymph node. **b** Fine needle aspiration of this lymph node was performed during EUS examination using EchoTip Ultra needle from Cook Medical, Bloomington, IN, USA

Fig. 2 A flow diagram showing the number of patients undergoing endoscopic ultrasonography (EUS)-guided fine needle aspiration cytology (FNAC) of the enlarged lymph nodes (LN) and outcome of further investigations



Chi-squared test was used to assess the differences in proportions between the paired groups. Kappa statistics were used to calculate absolute agreement between Gene Xpert, cytopathology, AFB smear, and conventional culture. Two-sided *p*-values of <0.05 were considered significant. Epi-info software (Centre for Disease Control, Atlanta, GA, USA) was used for analyzing the data.

Results

A total of 72 patients who underwent EUS-guided FNAC during this period were retrospectively reviewed (summarized in the consort diagram; Fig. 2). Of 31 patients with a diagnosis of TB (median age 33.5 years, IQR 21–66, 18, 58% female), clinical presentations were fever (28, 90%), dysphagia (2, 7%), and recurrent subacute intestinal obstruction (1, 3%). The demographic and clinical characteristics are given in Table 1.

Eleven (33%) patients had lymphadenopathy limited to a single station, and 20 (67%) had involvement of multiple stations. Periportal lymph nodes (15, 50%) were most commonly

sampled, followed by sub-carinal (7, 21%), peripancreatic (5, 15%), celiac (3, 11%), and paraesophageal lymph nodes (1, 3%) (Fig. 1). No complication occurred during the procedures. Twenty-four (77%) patients had either granulomatous or necrotizing lymphadenitis consistent with TB cytology (grade I, II, III) (Fig. 3).

Gene Xpert was more sensitive (30, 97%) than cytology (24, 77%; OR 8.8, 95% confidence interval [CI] 1.0–76.9; *p* = 0.05); AFB smears (12, 39%; OR 50, 95% CI 5.9–420.4; *p* = 0.00001); and conventional culture (4, 13%; OR 188.5, 95% CI 19.7–1796.3; *p* = 0.0000).

The true positive rate was compared between Gene Xpert and cytology using McNemar's exact test. There was a 20% difference in the true positive rate between Gene Xpert and cytology that was statistically significant (*p* = 0.04) (Table 2). Besides, there was a significant difference in the proportion of positive tests between Gene Xpert and others, i.e. conventional culture (97% vs. 13%, *p* < 0.001) and AFB smear (97% vs. 39%, *p* < 0.001).

All the patients responded to ATT. Only two patients developed anti-tubercular drug-induced hepatitis, which was successfully managed by modified anti-tubercular drugs.

Table 1 Baseline characteristics of patients included in the study (*n* = 31)

Male:female	13:18		
Median age (year)	33.5 (21–66)		
Variables		Frequency	Percentage
Sampling lymph node	Periportal	15	50
	Subcarinal	7	21
	Peripancreatic	5	14
	Celiac	3	11
	Paraesophageal	1	3
Cytology	I, II, III	24	77
	IV, V	7	23
Rifampicin sensitive (those showing Xpert positive)	Yes	30 (30)	100
Anti-tubercular drug regimen	Conventional (4 drugs)	29	94
	Modified	2	6

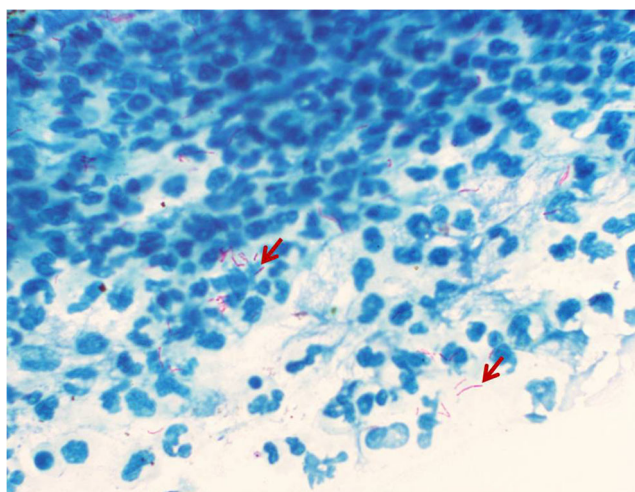


Fig. 3 Acid-fast bacilli (shown by arrow) amidst histiocytes, lymphocytes, neutrophilic polymorphs, and few epithelioid cells in endoscopic ultrasound-guided fine needle aspiration sample (Ziehl-Neelsen [ZN] stain; magnification $\times 1000$ oil immersion)

Discussion

Lymphadenitis in mediastinal and intra-abdominal regions is a common form of extra-pulmonary TB, and most of these patients have no active lung involvement [2]. EUS-guided cytology, in conjunction with demonstration of AFB on Ziehl-Neelsen (ZN) staining and conventional mycobacterial culture, may yield good results. However, limitations of traditional methods are low sensitivity, absence of species confirmation, long processing times, and lack of information about drug resistance [3]. Moreover, conventional microbiological culture and drug susceptibility testing may not be readily available and usually take 6 weeks or longer [16].

EUS features suggestive of tubercular lymphadenopathy include diffuse or patchy anechoic/hypoechoic echotexture, conglomeration, calcifications, sharply demarcated margins, large size, purulent aspirate, and cytopathological presence of necrosis/granuloma [17]. Gene Xpert MTB/RIF has been suc-

cessfully used to evaluate various extrapulmonary specimens, including urine, stool, and mediastinal lymph nodes. However, it has not been widely evaluated in EUS-guided intra-abdominal lymph nodal aspirates [13, 14, 18, 19]. In this retrospective analysis, we found that Gene Xpert's sensitivity was greater in tubercular lymph nodal aspirate (97%), reaching up to 100% when combined with cytology. Gene Xpert sensitivity for definite TB at 97% was consistent with sensitivities reported in the literature of 97% to 100% in respiratory and non-respiratory samples. The ability of the Gene Xpert test to simultaneously detect *Mycobacterium tuberculosis* and rifampicin resistance directly from FNA specimens is an added advantage over the culture and microscopy of smear for AFB [15, 18].

EUS-guided FNA of subcarinal and intra-abdominal lymph nodes is increasingly being used as a tool for microbiological diagnosis. In this analysis, it was observed that even a single Gene Xpert assay in isolation could improve sensitivity as compared to cytopathology and microscopy alone. In conjunction with supporting features (including cytology), Gene Xpert increased sensitivity and facilitated rapid diagnosis of suspected TB. Although rapid detection of rifampicin resistance led to immediate treatment changes (several weeks ahead of culture), the Xpert system's inability to identify all *rpoB* mutations highlights the need for alternative sequence-based detection methods.

In conclusion, this study revealed that the Xpert MTB/RIF test is a useful tool for the detection of *Mycobacterium tuberculosis* with high sensitivity on EUS-guided FNA and superior performance compared to cytology and smear microscopy. Besides improved sensitivity, the Xpert test is able to identify rifampicin resistance. This test is an easy and suitable method to be used in TB lymph nodal aspirates, and its implementation could significantly improve the rapid diagnosis of TB lymphadenitis. However, due to negligible number of negative cases, specificity and the negative predictive value could not be calculated. The effect of small sample size may have an impact on the results of the kappa statistics and McNemar test. The interesting results seen in the current study open the scope of studies with a larger sample size.

Table 2 Diagnostic accuracy of the tests with respect to clinical response

Tests	Clinical response (gold standard)		
	Sensitivity (TPR)	False negative rate	<i>p</i> -value
Gene Xpert (positive) (<i>n</i> =30/31)	30/31 (97%)	1/31 (3%)	<0.001
Cytology (positive, i.e. I, II, III) (<i>n</i> =24/31)	24/31 (77%)	7/31 (23%)	0.002
AFB smear (positive) (<i>n</i> =12/31)	12/31 (39%)	19/31 (61%)	0.25
Conventional culture (positive) (<i>n</i> =4/31)	4/31 (13%)	26/31 (87%)	<0.001
McNemar test: (Gene Xpert vs. Cytology, <i>p</i> =0.04)			

TPR true positive rate, AFB acid-fast bacillus

Compliance with ethical standards

Conflict of interest SM, HKN, NM, AK, VAS, NK, RM, and UCG declare that they have no conflict of interest.

Ethical issues All procedures performed in work were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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References

- Sharma SK, Mohan A. Multidrug-resistant tuberculosis. *Indian J Med Res.* 2004;120:354–76.
- Golden MP, Vikram HR. Extrapulmonary tuberculosis: an overview. *Am Fam Physician.* 2005;72:1761–8.
- Bezabih M, Mariam DW, Selassie SG. Fine needle aspiration cytology of suspected tuberculous lymphadenitis. *Cytopathology.* 2002;13:284–90.
- Shah SR, Shenai S, Desai DC, Joshi A, Abraham P, Rodrigues C. Comparison of Mycobacterium tuberculosis culture using liquid culture medium and Lowenstein Jensen medium in abdominal tuberculosis. *Indian J Gastroenterol.* 2010;29:237–9.
- Yasri S, Wiwanitkit V. Acid-fast bacilli culture positivity and drug resistance in abdominal tuberculosis. *Indian J Gastroenterol.* 2015;34:184.
- Samant H, Desai D, Abraham P, et al. Acid-fast bacilli culture positivity and drug resistance in abdominal tuberculosis in Mumbai, India. *Indian J Gastroenterol.* 2014;33:414–9.
- WHO 12 August 2010, accession date. WHO endorses new rapid tuberculosis test. . WHO, Geneva, Switzerland. 2010.
- World Health Organization [WHO]. Rapid implementation of the Xpert MTB / RIF diagnostic test: technical and operational “How-to”; practical considerations. Geneva; 2011.
- World Health Organization. Xpert MTB/RIF implementation manual: technical and operational ‘how-to’; practical considerations. 2014.
- Boehme CC, Nabeta P, Hilleman D, et al. Rapid molecular detection of tuberculosis and rifampin resistance. *N Engl J Med.* 2010;363:1005–15.
- Gandhi NR, Shah NS, Andrews JR, et al. HIV coinfection in multidrug- and extensively drug-resistant tuberculosis results in high early mortality. *Am J Respir Crit Care Med.* 2010;181:80–6.
- Boehme CC, Nicol MP, Nabeta P, et al. Feasibility, diagnostic accuracy, and effectiveness of decentralised use of the Xpert MTB/RIF test for diagnosis of tuberculosis and multidrug resistance: a multicentre implementation study. *Lancet.* 2011;377:1495–505.
- Blakemore R, Story E, Helb D, et al. Evaluation of the analytical performance of the Xpert MTB/RIF assay. *J Clin Microbiol.* 2010;48:2495–501.
- Denkinger CM, Schumacher SG, Boehme CC, Dendukuri N, Pai M, Steingart KR. Xpert MTB/RIF assay for the diagnosis of extrapulmonary tuberculosis: a systematic review and meta-analysis. *Eur Respir J.* 2014;44:435–46.
- Penz E, Boffa J, Roberts DJ, et al. Diagnostic accuracy of the Xpert(R) MTB/RIF assay for extra-pulmonary tuberculosis: a meta-analysis. *Int J Tuberc Lung Dis.* 2015;19:278–84, i-iii.
- Mistry Y, Ninama GL, Mistry K, et al. Efficacy of fine needle aspiration cytology, Ziehl-Neelsen stain and culture (Bactec) in diagnosis of tuberculosis lymphadenitis. *Natl J Med Res.* 2012;2:77–80.
- Bodh V, Choudhary NS, Puri R, et al. Endoscopic ultrasound characteristics of tubercular lymphadenopathy in comparison to reactive lymph nodes. *Indian J Gastroenterol.* 2016;35:55–9.
- Ligthelm LJ, Nicol MP, Hoek KG, et al. Xpert MTB/RIF for rapid diagnosis of tuberculous lymphadenitis from fine-needle-aspiration biopsy specimens. *J Clin Microbiol.* 2011;49:3967–70.
- Bera C, Michael JS, Burad D, et al. Tissue Xpert MTB/Rif assay is of limited use in diagnosing peritoneal tuberculosis in patients with exudative ascites. *Indian J Gastroenterol.* 2015;34:395–8.

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